Macular Degeneration, Early-onset (FBN2-related): A Comprehensive Disease Characterization

Disease: Macular Degeneration, Early-onset (EOMD) **MONDO ID:** MONDO:0014501 · **OMIM:** 616118 · **Causal gene: FBN2** (fibrillin-2) **Category:** Mendelian, autosomal dominant

Summary

Early-onset macular degeneration (EOMD; MONDO:0014501, OMIM 616118) is a rare autosomal-dominant Mendelian maculopathy caused by heterozygous variants in *FBN2*, the gene encoding fibrillin-2 — a large cysteine-rich glycoprotein of the elastin-rich extracellular matrix (ECM). The disease was defined molecularly by Ratnapriya et al. (2014), who used whole-exome sequencing of a two-generation family with dominant maculopathy to identify the index variant **p.Glu1144Lys** in *FBN2* (PMID: 24899048). The same study showed that *FBN2* sits at the interface of Mendelian and complex disease: a common non-synonymous *FBN2* variant (rs154001, p.Val965Ile) is also modestly associated with age-related macular degeneration (AMD), making *FBN2* an "allelic series" gene that links a rare early-onset dominant maculopathy to the far more common polygenic AMD.

Mechanistically, EOMD is best understood as a **Bruch's-membrane fibrillinopathy**. Fibrillin-2 localizes to the central elastic lamina of Bruch's membrane (BrM), the pentalaminar ECM sheet separating the retinal pigment epithelium (RPE) from the choriocapillaris. Loss or dysfunction of fibrillin-2 has two coupled consequences: (1) it weakens the structural elastic scaffold of BrM, and (2) it dysregulates the bioavailability of sequestered TGF-β/BMP growth factors, because fibrillins normally sequester these latent complexes in the ECM. The downstream cascade — elastin/microfibril breakdown, drusen accumulation, complement activation, elastin-derived-peptide-driven inflammation and angiogenesis, and ultimately choroidal neovascularization (CNV) — recapitulates in an accelerated, genetically-determined form the central ECM pathology of AMD.

There is **no gene-specific cure**. Clinical management is supportive and mirrors AMD care: anti-VEGF therapy (ranibizumab, aflibercept, brolucizumab, faricimab) for the neovascular (wet) complication, low-vision rehabilitation, and genetic counseling for an autosomal-dominant trait with apparently incomplete/age-dependent penetrance. Two mechanism-directed strategies are promising at the preclinical stage: **FBN2 recombinant-protein replacement** (rescued retinopathy in a retina-specific *Fbn2* knock-down mouse; PMID: 37100863) and **elastase inhibition** to preserve BrM elastin (PMID: 37174708).

Disease Information (Section 1)

Overview. EOMD is a rare inherited macular dystrophy with onset earlier than typical AMD, characterized by drusen-like deposits at the level of Bruch's membrane/RPE, progressive macular degeneration, and risk of choroidal neovascularization leading to central vision loss. It is caused by heterozygous variants in **FBN2 (fibrillin-2)**, an elastin-associated ECM glycoprotein.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0014501
OMIM	616118 (Macular degeneration, early-onset)
Gene	FBN2 (HGNC:3604; NCBI Gene 2201; Ensembl ENSG00000138829; chromosome 5q23.3)
Protein (UniProt)	P35556 (FBN2_HUMAN)
MeSH	Macular Degeneration (D008268) — no EOMD-specific MeSH term
ICD-10	H35.3 (degeneration of macula and posterior pole) — non-specific

Synonyms / alternative names: early-onset macular degeneration; EOMD; FBN2-related maculopathy; macular degeneration, early-onset, autosomal dominant.

Information source. The knowledge base entry is derived from **aggregated disease-level resources** (OMIM, primary literature) and a small number of individual pedigrees/case series rather than large EHR cohorts. The defining evidence comes from a single index family plus mutation screening of additional maculopathy patients (PMID: 24899048).

Etiology (Section 2)

Primary cause — genetic. EOMD is a Mendelian, autosomal-dominant disorder. The primary cause is a heterozygous pathogenic variant in **FBN2**. The index variant is a missense change, **p.Glu1144Lys**, identified by whole-exome sequencing:

> *"we performed whole exome sequencing in four individuals of a two-generation family with autosomal dominant maculopathy and identified a rare variant p.Glu1144Lys in Fibrillin 2 (FBN2), a glycoprotein of the elastin-rich extracellular matrix (ECM)"* — PMID: 24899048

Screening of 192 additional maculopathy patients revealed further rare *FBN2* variants predicted to disrupt function, supporting an allelic series.

Genetic risk factors (complex disease). A common *FBN2* non-synonymous variant **rs154001 (p.Val965Ile)** showed suggestive association with AMD (OR = 1.10; P = 3.79×10^{-5}) in 10,337 cases vs 11,174 controls, demonstrating that the same locus contributes to multifactorial AMD risk:

> *"We detect suggestive association of a common FBN2 non-synonymous variant, rs154001 (p.Val965Ile) with AMD in 10 337 cases and 11 174 controls (OR = 1.10; P-value = 3.79 × 10(-5))"* — PMID: 24899048

Environmental risk factors. No EOMD-specific environmental triggers have been established. By analogy with AMD, aging, oxidative stress, and smoking plausibly accelerate BrM ECM degradation, but direct evidence in FBN2-EOMD is **not available**. Because EOMD is monogenic and early-onset, environmental factors are expected to act as modifiers rather than primary causes.

Protective factors. No genetic or environmental protective factors specific to EOMD have been characterized. (Not available.)

Gene–environment interactions. Not directly studied in EOMD. The dual role of *FBN2* (rare dominant variant → EOMD; common variant → AMD susceptibility) suggests the gene's ECM/growth-factor function is a convergence point where genetic load and age-related/environmental ECM stress combine. This remains inferred.

Phenotypes (Section 3)

EOMD phenotypes are ophthalmologic and centered on the macula. Because the molecularly-defined cohort is small, frequencies are qualitative.

Phenotype	Type	HPO term (suggested)	Characteristics
Macular degeneration	Clinical sign	HP:0000608 (Macular degeneration)	Early-onset (earlier than typical AMD); progressive
Drusen / sub-RPE deposits	Clinical sign (imaging)	HP:0011510 (Macular drusen)	Hallmark BrM-level deposits; early sign
Reduced visual acuity	Symptom	HP:0000572 (Visual impairment)	Variable; progressive central vision loss
Choroidal neovascularization	Clinical sign (complication)	HP:0011506 / CNV	Sight-threatening wet complication
Central scotoma	Symptom	HP:0000592 (Central scotoma)	Corresponds to macular lesions
Metamorphopsia	Symptom	—	Distortion typical of maculopathy
Reduced ERG amplitude (model)	Laboratory/functional	HP:0000512 (Abnormal electroretinogram)	Shown in Fbn2 knock-down mouse (PMID: 37100863)

Age of onset: Earlier than age-related AMD — childhood-to-adult rather than geriatric; exact distribution is not well-defined given the small cohort. **Severity:** Variable, ranging from drusen-only to chorioretinal involvement with CNV. **Progression:** Progressive. **Quality-of-life impact:** Central vision loss impairs reading, writing, driving, face recognition, and stereoscopic

vision — the same functional losses that make AMD the leading cause of irreversible central blindness in older adults (PMID: 24938626). EOMD-specific QoL instruments (EQ-5D/SF-36) have not been reported. (Not available.)

Genetic / Molecular Information (Section 4)

Causal gene. *FBN2* (fibrillin-2), HGNC:3604, chromosome 5q23.3, UniProt P35556. FBN2 is a ~2,900-residue modular glycoprotein built mainly of calcium-binding EGF-like domains and TB (TGF-β-binding/8-cysteine) domains that templates microfibril and elastic-fiber assembly.

Pathogenic variants.

Variant	Type	Disease	Notes
p.Glu1144Lys (NM_001999)	Missense	EOMD (index)	Segregated in two-generation pedigree (PMID: 24899048)
Additional rare FBN2 variants	Predicted damaging	EOMD/ maculopathy	Found by screening 192 patients
rs154001 (p.Val965Ile)	Missense, common	AMD susceptibility	OR 1.10, P = 3.79×10^{-5}

Variant classification: The index p.Glu1144Lys is reported as a rare, segregating, disease-causing variant; by ACMG/AMP framing it draws support from segregation (PP1), rarity (PM2), and in-silico damage prediction (PP3). Many *FBN2* variants elsewhere in ClinVar relate to **congenital contractural arachnodactyly (CCA / Beals-Hecht syndrome)** rather than EOMD (see below), so curation must distinguish the ocular from the skeletal phenotype.

Allele frequency: The index variant and EOMD-associated variants are rare in population databases (gnomAD); rs154001 is common.

Somatic vs germline: Germline. No somatic involvement.

Functional consequence: Loss / reduction of fibrillin-2 function. Retinal FBN2 protein is reduced in both AMD and EOMD patients, and a loss-of-function mouse recapitulates retinopathy that is rescued by FBN2 protein — consistent with a **loss-of-function / haploinsufficiency-like** mechanism for the ocular phenotype:

> *"the retinal protein expression of FBN2 was reduced in patients with AMD and EOMD"* — PMID: 37100863

Genotype–phenotype note — allelic disorders. *FBN2* is pleiotropic. Variants in the **central region (classically exons 24–35)** of *FBN2* typically cause **congenital contractural arachnodactyly (CCA/Beals syndrome)**, an autosomal-dominant connective-tissue disorder with arachnodactyly, camptodactyly, joint contractures, crumpled ears, and scoliosis (PMID: 42299845; PMID: 39911746). Variants **outside the central region** are linked to thoracic aortic disease and to short stature/brachydactyly/joint-contracture phenotypes (PMID: 40406865; PMID: 40199564). The EOMD-associated variants constitute a distinct, ocular-predominant arm of the *FBN2* allelic spectrum.

Modifier genes: Not formally defined for EOMD. Candidate modifiers by analogy with AMD ECM biology include *EFEMP1*, *TIMP3*, *FBLN5*, complement genes (*CFH*), *ARMS2/HTRA1*, and ADAMTS proteases that process fibrillins (PMID: 24938626).

Epigenetic information / chromosomal abnormalities: No EOMD-specific DNA-methylation, histone-modification, or large-scale chromosomal findings reported. (Not available / not applicable.)

Environmental Information (Section 5)

- **Environmental factors / toxins:** None established as causal for EOMD (monogenic disease). (Not available.)
- **Lifestyle factors:** Not specifically studied in EOMD; smoking and oxidative stress are AMD risk factors that may plausibly accelerate shared BrM pathology but are unproven here.
- **Infectious agents:** Not applicable. EOMD is a genetic, non-infectious ECM disorder.

Mechanism / Pathophysiology (Section 6)

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Heterozygous *FBN2* variant (e.g., p.Glu1144Lys)** → leads to reduced/dysfunctional fibrillin-2 protein in the retina/Bruch's membrane (demonstrated: reduced FBN2 protein in patient retina; PMID: 37100863).
2. **Reduced functional fibrillin-2** → results in defective microfibril/elastic-fiber assembly in the **central elastic lamina of Bruch's membrane**, where FBN2 normally resides (demonstrated localization: PMID: 24899048).
3. *Branch A — structural:* **Weakened elastic lamina** → reduces the integrity of the BrM molecular sieve → promotes abnormal deposition (drusen) and loss of BrM barrier function (inferred from AMD BrM biology; PMID: 19747980).
4. *Branch B — signaling:* **Loss of fibrillin-2's sequestration of latent TGF-β/BMP complexes** → results in dysregulated TGF-β/BMP bioavailability and signaling (established fibrillin function; PMID: 19513754, PMID: 17708531).
5. **Elastin/microfibril breakdown** (aggravated by excess elastase and reduced protease inhibition) → generates elastin-derived peptides (EDPs) and anti-elastin antibodies → drives inflammatory and angiogenic responses (demonstrated associations; PMID: 35798060, PMID: 37174708).
6. **Drusen + complement cascade activation** → represent the earliest degenerative signs at the RPE–BrM interface (inferred from AMD; PMID: 19747980).
7. **Reduced BrM elastin integrity + elevated VEGF** → leads to **choroidal neovascularization (CNV)** in the macula (demonstrated spatial correspondence; PMID: 37174708).
8. **Macular degeneration / CNV** → results in central scotoma, metamorphopsia and progressive central vision loss (clinical manifestation).

Detail by category

Molecular pathways. Central pathway is **TGF-β/BMP superfamily signaling**, regulated by fibrillin microfibrils. Fibrillins sequester latent TGF-β and BMP in the ECM:

> *"the instructive role reflects the ability of fibrillins to sequester transforming growth factor beta (TGFbeta) and bone morphogenetic protein (BMP) complexes in the extracellular matrix"* — PMID: 19513754

> *"fibrillin-1 and fibrillin-2 mutations perturb signaling events mediated by TGF-beta superfamily members"* — PMID: 17708531

ADAMTS proteases (ADAMTS6/10, ADAMTSL2) that process fibrillins and LTBPs further couple microfibril turnover to TGF-β activation, YAP/TAZ mechanotransduction and downstream angiogenic signaling (PMID: 36368447; PMID: 30060141) — suggested GO terms: **GO:0007179 (transforming growth factor beta receptor signaling pathway)**, **GO:0030509 (BMP signaling pathway)**.

Cellular processes. Inflammation (complement activation), angiogenesis (CNV), and RPE/endothelial ECM remodeling. GO:0006954 (inflammatory response), GO:0001525 (angiogenesis), GO:0030198 (extracellular matrix organization).

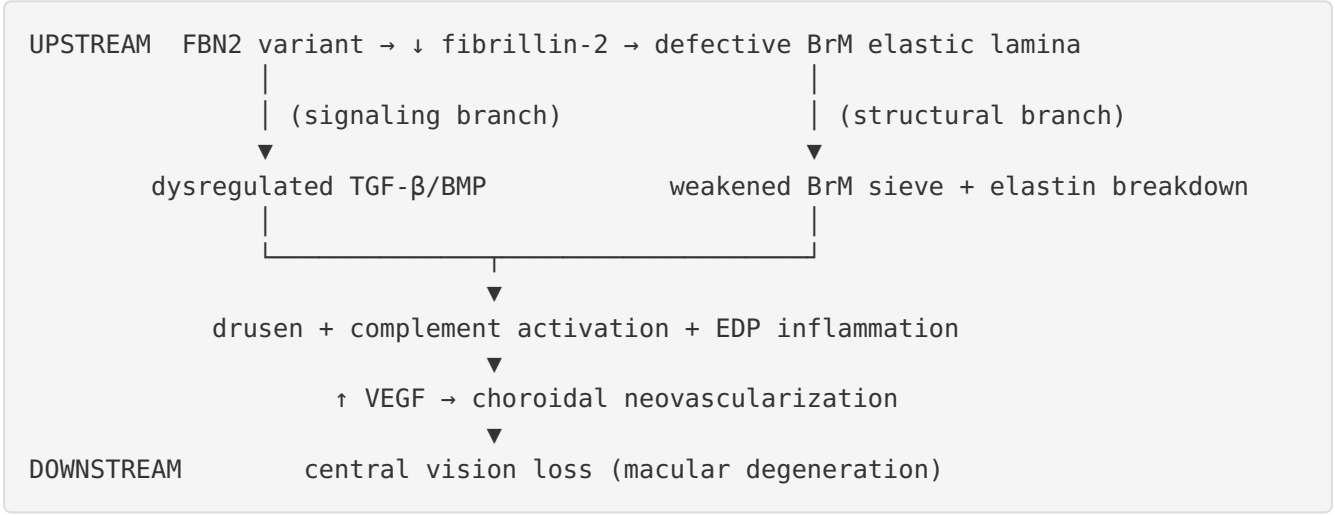
Protein dysfunction. Loss/reduction of fibrillin-2 function → impaired elastic-fiber template assembly (GO:0048251, elastic fiber assembly) and impaired growth-factor sequestration.

Tissue damage mechanisms. Oxidative/proteolytic degradation of BrM elastin; elastase activity degrades elastin and drives VEGF; AMD donor BrM/choroid shows lower alpha-1 antitrypsin (a serine-protease inhibitor), tilting the protease/anti-protease balance toward elastin loss (PMID: 40210893).

Immune involvement. Complement cascade activation at drusen; EDP-driven inflammation and anti-elastin antibodies (PMID: 35798060).

Cell types (suggested CL terms): retinal pigment epithelial cell (CL:0002586), choroidal/blood vessel endothelial cell (CL:0000071), fibroblast-like ECM-producing cells.

Upstream vs downstream



Anatomical Structures Affected (Section 7)

Level	Structure	Ontology (suggested)
Organ	Eye; retina (macula)	UBERON:0000970 (eye); UBERON:0001782 (macula lutea)
Primary site	**Bruch's membrane** (central elastic lamina)	Bruch's membrane
Adjacent	Retinal pigment epithelium	RPE
Adjacent	Choroid / choriocapillaris	UBERON:0001776 (choroid)
Tissue	Elastin/collagen-rich ECM (connective)	—
Cell	RPE cell; choroidal endothelial cell	CL:0002586; CL:0000071
Subcellular	Extracellular matrix (secreted)	GO:0031012 (extracellular matrix)

Bruch's membrane is the defining locus:

> *"BM is an elastin- and collagen-rich extracellular matrix that acts as a molecular sieve"* — PMID: 19747980

Lateralization: Bilateral (hereditary maculopathy), though asymmetric severity/CNV onset is common.

Temporal Development (Section 8)

- **Onset:** Earlier than age-related AMD (hence "early-onset"); insidious, chronic.
- **Stages:** (1) drusen-only/sub-RPE deposits → (2) macular degenerative change → (3) choroidal neovascularization (wet, sight-threatening). This parallels the dry→wet AMD progression.

- **Progression rate:** Slow and progressive, with variable expressivity (drusen-only to chorioretinal involvement, as seen in dominant early-onset maculopathy pedigrees; PMID: 17302207).
- **Course:** Chronic, lifelong, progressive; no spontaneous remission.
- **Critical period for intervention:** The window before/at CNV onset — anti-VEGF therapy is most beneficial when neovascular activity is treated promptly.

Inheritance and Population (Section 9)

- **Inheritance pattern:** Autosomal dominant.

> *"Sanger sequencing validated the segregation of this variant in the complete pedigree, including two additional affected and one unaffected individual"* — PMID: 24899048

- **Penetrance:** Apparently incomplete / age-dependent — the index pedigree included an unaffected carrier.
- **Expressivity:** Variable (drusen-only to CNV).
- **Epidemiology:** EOMD is very rare; precise prevalence/incidence are **not established** (defined by individual pedigrees/case series rather than registries).
- **Genetic anticipation / germline mosaicism / founder effects:** Not reported for EOMD. (Note: low-level parental mosaicism has been documented for *FBN2* in the CCA phenotype, PMID: 42299845, but not for EOMD.)
- **Consanguinity:** Not relevant (dominant).
- **Carrier frequency:** Not applicable in the recessive sense; disease alleles are rare.
- **Population demographics / sex ratio / geographic distribution:** No ethnic predilection or sex bias established; the common AMD-associated rs154001 is distributed broadly. (EOMD-specific data not available.)

Diagnostics (Section 10)

Clinical/ophthalmic workup. - **Imaging:** Color fundus photography, spectral-domain OCT (sub-RPE vs sub-retinal deposits), fundus autofluorescence, fluorescein angiography (FA) and indocyanine-green angiography (ICGA) for CNV, OCT-angiography. Multimodal imaging characterizes drusen and detects CNV (PMID: 22496012; PMID: 42132463). - **Electrophysiology:** Full-field ERG (often normal), multifocal ERG (reduced macular amplitude/delayed implicit time), EOG — useful to localize dysfunction to the macula (PMID: 17302207).

Genetic testing (confirmatory). Diagnosis is confirmed by identifying a pathogenic *FBN2* variant via: - **Single-gene *FBN2* sequencing**, or - **Inherited-retinal-disease / maculopathy gene panels**, or - **Whole-exome sequencing (WES)** — the method used in the original discovery (PMID: 24899048).

WGS and chromosomal microarray add little beyond targeted sequencing for a point-variant disorder; karyotyping/FISH/mtDNA/repeat-expansion testing are **not applicable**.

Biomarkers (research-stage). Elevated serum elastin-derived peptides (EDPs) and anti-elastin antibodies accompany reduced BrM elastin integrity in AMD and may be relevant to EOMD (PMID: 35798060). Reduced retinal FBN2 protein is a tissue-level marker (PMID: 37100863). None are clinically validated for EOMD.

Differential diagnosis. Other autosomal-dominant early-onset maculopathies, distinguished genetically:

Condition	Gene	Distinguishing feature
Malattia Leventinese / Doyne honeycomb dystrophy	*EFEMP1* (p.Arg345Trp)	Radial/honeycomb drusen (PMID: 41971199; PMID: 42720932)
North Carolina macular dystrophy (phenocopy)	*PRDM13*/MCDR1 region	Early non-progressive macular lesions (PMID: 17302207)
Sorsby fundus dystrophy	*TIMP3*	Early CNV, night blindness
Pattern dystrophy / Best disease	*PRPH2* / *BEST1*	Vitelliform lesions, abnormal EOG
Stargardt disease	*ABCA4*	Flecks, usually recessive

> *"Malattia Leventinese is a rare autosomal dominant macular dystrophy caused by a single"* — PMID: 41971199

Screening. Cascade genetic testing of at-risk relatives once the familial *FBN2* variant is known. No population newborn/carrier screening (rare dominant disorder).

Outcome / Prognosis (Section 11)

- **Survival/mortality:** EOMD is **not life-limiting**; it does not affect survival. (The ocular-predominant *FBN2* phenotype is distinct from aortic/connective-tissue *FBN2* phenotypes that carry systemic risk.)
- **Morbidity/disability:** Principal burden is progressive central vision loss impairing reading, driving, and face recognition — the functional disability of macular disease.
- **Disease course:** Chronic progressive; CNV is the major sight-threatening complication.
- **Prognostic factors (CNV/visual outcome, extrapolated from neovascular macular disease):** younger age, better baseline visual acuity, smaller hemorrhage size, and thicker subfoveal choroid predict better visual outcomes in submacular hemorrhage from neovascular disease (PMID: 42177934). Foveal involvement predicts worse acuity in peripapillary CNV (PMID: 42132463).
- **Recovery potential:** No recovery of lost photoreceptors; anti-VEGF can stabilize/partially improve acuity in the wet phase.

Treatment (Section 12)

There is **no gene-specific approved therapy**. Management mirrors AMD.

Pharmacotherapy — anti-VEGF (standard of care for the neovascular/wet complication). Intravitreal **ranibizumab, aflibercept, brolucizumab, faricimab** (and biosimilars) reduce central macular thickness and stabilize vision (NCIT: *Anti-VEGF Therapy*; *Ranibizumab* C48094; *Aflibercept* C70846). Evidence base: aflibercept 8 mg improves anatomy in refractory

neovascular AMD (PMID: 42026508); treat-and-extend regimens with longer-half-life agents reduce treatment burden (PMID: 42054086); ranibizumab biosimilars are non-inferior and improve access (PMID: 42054089).

Advanced / experimental (preclinical for EOMD specifically): - **FBN2 protein replacement:** Intravitreal recombinant fbn2 rescued retinopathy in a retina-specific *Fbn2* knock-down mouse —

> *"After additional and repeated application of fbn2 recombinant protein, the retinopathy improved with an increase in retinal thickness and ERG amplitude"* — PMID: 37100863

- **Elastase inhibition:** Preserving BrM elastin is a therapeutic strategy in mouse and human AMD models (PMID: 37174708); alpha-1 antitrypsin augmentation is a related protease-inhibition approach (PMID: 40210893).
- **Gene therapy / RNA therapy:** Conceptually attractive (AAV gene supplementation or editing) but **not yet reported** for *FBN2*-EOMD; the large *FBN2* cDNA (~8.6 kb) exceeds standard AAV capacity, a practical hurdle.

Supportive/rehabilitative: Low-vision aids, Amsler-grid self-monitoring, occupational therapy; AREDS-type antioxidant supplementation is of unproven benefit specifically in EOMD.

Pharmacogenomics: No EOMD-specific pharmacogenomic guidance. (Not available.)

Prevention (Section 13)

- **Primary prevention:** Not possible for the genetic cause; general ocular-health and anti-oxidant/anti-smoking measures are reasonable but unproven in EOMD.
- **Secondary prevention:** Early detection via ophthalmic surveillance (OCT, Amsler grid) in known carriers to catch CNV promptly.
- **Tertiary prevention:** Prompt anti-VEGF for CNV to prevent irreversible vision loss.
- **Genetic counseling:** Central — autosomal-dominant inheritance with 50% transmission risk and incomplete/age-dependent penetrance; cascade testing and reproductive options (prenatal/PGT) can be discussed once the familial variant is known.
- **Immunization / public-health / environmental interventions:** Not applicable.

Other Species / Natural Disease (Section 14)

- **Orthologous gene:** *Fbn2* (mouse NCBI Gene 14119; chromosome 18), highly conserved; fibrillin-2 function in elastic-fiber assembly and TGF-β/BMP regulation is evolutionarily conserved (PMID: 19513754).
- **Natural disease in other species:** No well-characterized naturally-occurring FBN2 maculopathy in companion animals/wildlife is documented in the reviewed literature. (Not available.)
- **Comparative biology:** The conserved fibrillin/TGF-β axis underlies multiple species' connective-tissue phenotypes; mouse *Fbn2* models are informative (below).
- **Zoonotic potential:** Not applicable (non-infectious genetic disease).

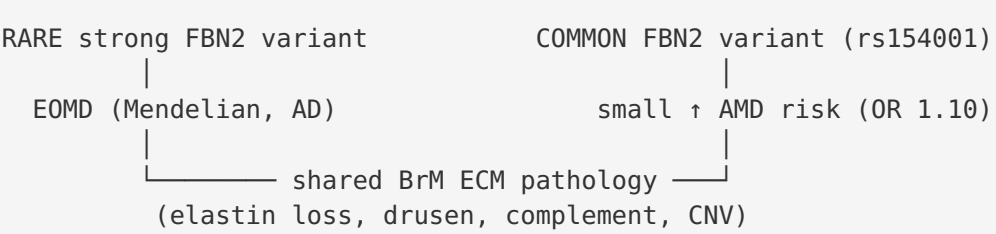
Model Organisms (Section 15)

Model	Type	Key features	Reference
Retina-specific *Fbn2* knock-down mouse (AAV-sh-fbn2)	Mammalian, induced genetic	Exudative retinopathy of deep retinal layers, reduced axial length, reduced ERG amplitudes; **rescued by intravitreal fbn2 recombinant protein**	PMID: 37100863
ADAMTS10 WMS mouse (CRISPR)	Mammalian	Perturbed fibrillin microfibrils, abundant fibrillin-2, altered BMP/GDF8 and SMAD1/5/8 vs p38/MAPK signaling	PMID: 30060141
FBN2-related connective-tissue models	Mammalian	Inform TGF-β/BMP dysregulation (systemic, not ocular)	PMID: 17708531

Phenotype recapitulation: The retina-specific *Fbn2* knock-down mouse reproduces the loss-of-function retinopathy and, importantly, demonstrates reversibility with protein replacement — strong support for a dosage-sensitive, treatable mechanism. **Limitations:** It is a knock-down (not the human missense allele), targets retina acutely, and does not model the slow, lifelong human BrM degeneration or dominant inheritance dynamics. **Resources:** MGI (*Fbn2*), IMPC/IMSR for additional alleles.

Mechanistic Model / Interpretation

EOMD is a **monogenic model of the ECM arm of AMD**. The unifying interpretation across all five confirmed findings is that **fibrillin-2 is a structural-and-signaling keystone of the Bruch's-membrane elastic lamina**, and that reducing its function tips the macular RPE–BrM–choroid complex toward the same degenerative endpoint reached more slowly in polygenic, age-related AMD. The two mechanistic branches (structural weakening of the elastic sieve; dysregulated TGF-β/BMP bioavailability) are not alternatives but co-occurring consequences of the same lesion, and they converge on drusen formation, complement/inflammatory activation, elastin-breakdown-driven angiogenesis, and CNV. The fact that **common *FBN2* variation modestly raises AMD risk while rare *FBN2* variation causes early-onset Mendelian disease** is the clearest evidence that this single ECM gene operates along a dosage/severity continuum.



Evidence Base

PMID	Role in this report	Evidence type
24899048	**Defining paper** — identifies FBN2 p.Glu1144Lys in EOMD family; BrM localization; rs154001 AMD association	Human genetics (WES + case-control)
37100863	Reduced FBN2 in AMD/EOMD retina; knock-down mouse retinopathy rescued by FBN2 protein	Human tissue + model organism
19513754	Fibrillins sequester TGF-β/BMP in ECM	Review / mechanism
17708531	FBN mutations perturb TGF-β superfamily signaling	Review / mechanism
19747980	BrM composition/function; drusen + complement as earliest AMD signs	Review
35798060	Reduced BrM elastin integrity; EDPs and anti-elastin antibodies elevated	Review
37174708	BrM elastin breakdown corresponds to CNV; elastase inhibition therapeutic	Human + mouse
40210893	Alpha-1 antitrypsin maintains BrM integrity; lower in AMD donors	Mouse + human donor
24938626	ECM/MMP-TIMP dysregulation in AMD BrM	Review
41971199, 42720932	EFEMP1 differential diagnosis	Human clinical
17302207	Dominant early-onset maculopathy phenotyping/differential	Human clinical
36368447, 30060141	ADAMTS–fibrillin–TGF-β coupling	In vitro / mouse
42026508, 42054086, 42054089, 42177934, 42132463	Anti-VEGF treatment evidence & CNV prognosis (extrapolated)	Human clinical
42299845, 39911746, 40406865, 40199564, 40436996	FBN2 allelic disorders (CCA/aortic) — genotype–phenotype context	Human clinical

Limitations and Knowledge Gaps

1. **Small human cohort.** EOMD is defined by essentially one index family plus mutation screening; prevalence, penetrance estimates, age-of-onset distribution, and sex ratio are not robustly established.
2. **Mechanism largely inferred from AMD.** Much of the BrM elastin/complement/CNV cascade is demonstrated in AMD and model systems, then extrapolated to EOMD; direct EOMD tissue studies are limited to reduced FBN2 protein.
3. **Variant-specific functional data sparse.** The precise molecular consequence of p.Glu1144Lys (loss-of-function vs dominant-negative vs altered TGF-β sequestration) is not fully resolved.
4. **No EOMD natural history or registry data**; no validated EOMD-specific biomarkers or QoL instruments.
5. **Therapeutics are preclinical** (protein replacement, elastase inhibition); the large *FBN2* cDNA complicates AAV gene therapy.
6. **Modifier genes and gene–environment interactions** remain hypothetical.

Proposed Follow-up Experiments / Actions

1. **Expand the FBN2-EOMD cohort** via inherited-retinal-disease registries and GeneMatcher to firm up penetrance, expressivity, and onset distribution, and to build an allelic map distinct from CCA/aortic FBN2 variants.
2. **Functional characterization of p.Glu1144Lys** in RPE/organoid and microfibril-assembly assays: quantify microfibril integrity, TGF-β/BMP activation (SMAD2/3 vs SMAD1/5/8), and whether the allele is loss-of-function or dominant-negative.
3. **Knock-in mouse (or iPSC-RPE)** carrying the human missense allele to model dominant inheritance and slow BrM degeneration more faithfully than the knock-down.
4. **Biomarker validation:** test serum EDPs, anti-elastin antibodies, and circulating/tissue FBN2 as EOMD diagnostic/prognostic markers.
5. **Therapeutic development:** advance FBN2 protein replacement and elastase/protease-inhibition (including alpha-1 antitrypsin augmentation) toward CNV-prevention endpoints; evaluate dual-AAV or non-viral delivery to overcome FBN2 size limits.

6. **TGF-β/BMP axis modulation:** test whether normalizing TGF-β bioavailability mitigates drusen/CNV in models, given the ADAMTS–fibrillin–TGF-β coupling.

Report compiled from 5 discovery iterations, 5 confirmed findings, and 26 reviewed papers. Evidence types are labeled (human clinical, model organism, in vitro, review/computational) throughout.
